Political complexity predicts the spread of ethnolinguistic groups

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Human languages show a remarkable degree of variation in the area they cover. However, the factors governing the distribution of human cultural groups such as languages are not well understood. While previous studies have examined the role of a number of environmental variables the importance of cultural factors has not been systematically addressed. Here we use a geographical information system (GIS) to integrate information about languages with environmental, ecological, and ethnographic data to test a number of hypotheses that have been proposed to explain the global distribution of languages. We show that the degree of political complexity and type of subsistence strategy exhibited by societies are important predictors of the area covered by a language. Political complexity is also strongly associated with the latitudinal gradient in language area, whereas subsistence strategy is not. We argue that a process of cultural group selection favoring more complex societies may have been important in shaping the present-day global distribution of language diversity.

cultural diversity | cultural evolution | cultural group selection | language diversity | latitudinal gradient

Human cultural groups are not evenly distributed across the earth's surface. For example, 235 languages are spoken in China, an area of 9.5 million km^2, whereas ≈1,000 languages are spoken on the island of New Guinea, a region less than a tenth of the size (1). Cultural and linguistic diversity seems broadly to follow a latitudinal gradient, with an increasing density of groups from the poles toward the equator (2) (Fig. 1*A*). A similar phenomenon in biological species richness is one of the best-attested patterns in ecology and has been commented on since at least the time of Alfred Russel Wallace (3). Despite over 30 hypotheses having been postulated to explain the latitudinal gradient in species richness, there is as yet no consensus (4). In contrast, relatively little attention has been paid to understanding the patterns of diversity seen in the distribution of human ethnolinguistic groups, despite the fact that such studies can reveal the factors that may affect the origin and maintenance of human cultures and cultural diversity (5, 6).

In general, areas of high language diversity will have languages that cover smaller areas than languages in regions of lower diversity (5). Previous work on North American languages (7) has shown that language area increases with increasing latitude (5). Fig. 1*B* shows that this is also the case for the languages of the Old World; as latitude increases the number of languages decreases and this is associated with an increase in the average area covered by those languages. While a latitudinal gradient in language area suggests an important role for environmental factors in determining the distribution of cultural groups the effect of cultural factors has not been investigated. As some factors may influence the area covered by a language but are not responsible for creating larger-scale patterns such as the latitudinal gradient, we ask 2 related questions: (*i*) Which environmental and cultural factors predict the area covered by a language? (*ii*) Do these factors also explain the latitudinal gradient in language area, and, by extension, language diversity? In this paper we integrate information from languages with ethnographic, environmental, and ecological data to test a number of hypotheses that could explain the distributions of languages.

Environmental Factors. Cultural group diversity has been found to correlate with a number of environmental variables (5, 8–15). There are a number of reasons why environmental factors may affect the area over which a language is spread. Geographical features such as mountains can create barriers to the movement of people and may lead to the isolation and divergence of cultural groups in a manner analogous to allopatric speciation (16); therefore, regions of topographic heterogeneity should contain smaller languages. Stepp and colleagues (11) argue that the mountainous environments on the island of New Guinea is important in maintaining high levels of both biological and cultural diversity in the region. Some authors (9, 17) have proposed that language areas reflect a response to environmental uncertainty: in riskier environments social networks extend over larger areas to buffer against local shortfalls in subsistence. After finding an association between the number of languages in a country and a measure of ecological risk, Nettle asserted that "no factor has been as strong or as general as ecological risk" (ref. 9, p. 94) in determining the distribution of languages. In an analysis of Australian Aboriginal tribal areas, Birdsell (18) offered another explanation arguing that environmental productivity was an important factor in determining the area of land required by groups to meet basic subsistence needs. He found that in regions of lower rainfall an approximately equal number of individuals [the "magic number" 500, (19)] were spread out over larger areas than those in the more productive regions of higher rainfall. Finally, the fact that regions of high biological and cultural diversity tend to overlap has led several authors to propose that the two are interdependent (20). The precise mechanism has not been made explicit, however, and the relationship between biological and cultural diversity may instead be indirect.

As many environmental variables also vary with latitude, spurious correlations with cultural diversity might emerge, particularly at the coarser grains of analysis used in previous studies (8). Our methods allow a fine-grained analysis to test which environmental factors are important in explaining the area a language covers.

Cultural Factors. Human societies exhibit a remarkable degree of variation in their social organization that may have consequences for the distribution of languages. Importantly, cultural differences between groups may allow some societies to expand at the expense of others (21). In this study we examine how 2 factors,

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ANTHROPOLOGY

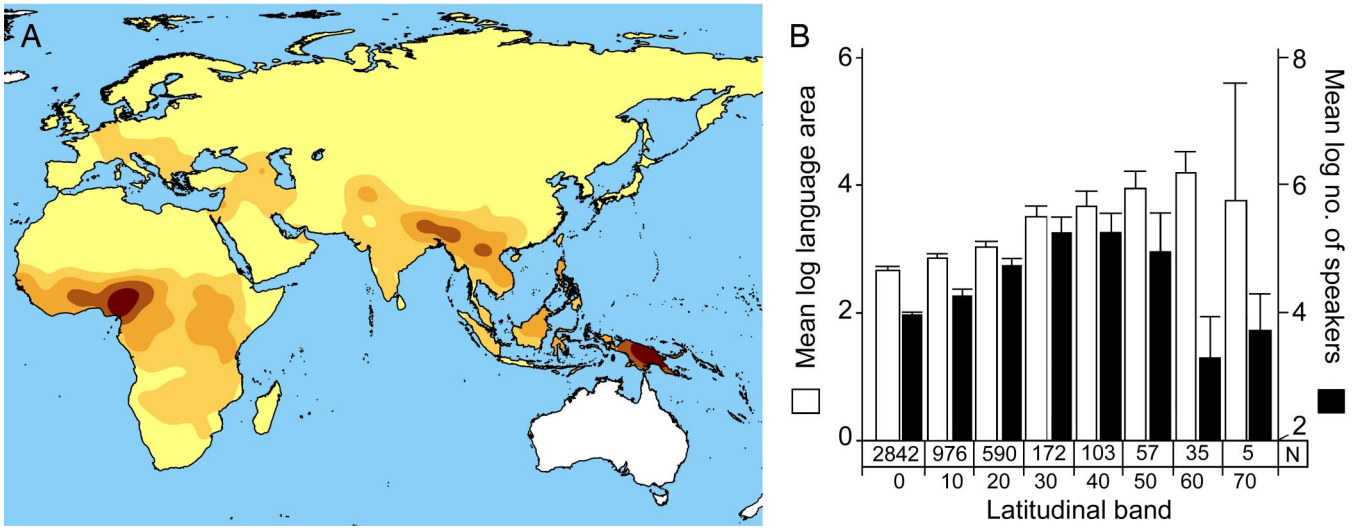


Fig. 1. Distribution of languages in the Old World. (*A*) Kernel density map of languages of the Old World (created in ArcGIS). Darker colors represent regions of higher language density. (*B*) Average log language area (black bars) increases with absolute latitude (*Lower* bound of 10° bands). *N* represents the number of languages at each band of latitude. Larger language areas are not simply the result of a relatively constant number of speakers being spread out over larger areas as the average number of speakers (white bars) also increases with increasing language area up to 60° of latitude.

subsistence strategy and political complexity, may affect language areas.

Subsistence strategies based on food production are able to support higher population densities than those based on hunting and gathering (22). This means people will not have to travel as far to meet their subsistence requirements, interact with others, exchange food and goods, and find mates (23). Hunter–gatherers and pastoralists therefore have to be more mobile to be able to find new food sources or places to graze their animals when there are no longer enough resources in the region they are inhabiting (9). Accordingly, all other things being equal, people with more mobile subsistence strategies (i.e., foraging and pastoralism) may spread their languages over larger areas than those with less mobile strategies (i.e., agriculture and fishing). Furthermore, if subsistence strategies are responsible for the latitudinal gradient seen in language diversity, then subsistence strategies associated with larger language areas should be more common at higher latitudes.

Political complexity is also a potentially important factor in the area a language covers that has largely been ignored in previous studies of ethnolinguistic diversity. Before the development of agriculture, archaeological evidence suggests that human societies were organized only at a very local level made up primarily of families or groups of families related by common descent (24). Since food production began, more politically "complex" forms of societies have emerged, that involve the integration and coordination of larger numbers of people (25, 26). Some of these societies exhibit hereditary inequalities between lineages and individuals, permanent offices of leadership, craft specialists, and professional warriors that are supported by others in the population. If groups do amalgamate, then cultural traits (such as language) may homogenize for such reasons as coercion, the need to reduce interaction costs (e.g., to allow effective communication), or prestige bias (21). As more politically complex societies can generally coordinate the actions of larger numbers of individuals more effectively (27, 28) they will generally have an advantage in competition with less complex groups and can expand at their expense or absorb them, and thus will spread their languages over wider areas. An example of this process can be seen in the spread of the Russian state, which originally was centered on Moscow and the surrounding area but began to expand eastwards in the 16th century, eventually reaching the Pacific Ocean. Indigenous groups were unable to prevent this expansion as they had no history of acting in a coordinated fashion with other groups and because of the disparity in military technology between themselves and the Russians (ref. 29, p.146). The Russian language today is spread over an area of more than 9 million km^2.

If political complexity has been an important force in determining the area over which a language is spread, then languages associated with more politically complex societies should have larger areas. Furthermore, if political complexity is important in creating the latitudinal gradient in language diversity, then more complex societies should be found at higher latitudes.

It should be stressed that "complexity" here refers only to the political organization of a society and can be indexed by the number of hierarchical jurisdictional levels in a society (30). Other aspects of social organization may of course show different patterns of complexity. For example, the kinship structures of the Aranda of Australia would be considered more complex than those of most European societies even though they are politically less complex under the definition given here (31). We also clearly recognize that environmental and cultural factors can be interrelated, and it is not our desire to set up a false dichotomy between "environment" and "culture." Human subsistence strategies, for example, undoubtedly reflect adaptations to the physical environment (32), and many cultural traits of societies may ultimately depend on environmental or geographical conditions (25). However, by assessing the relative contributions of environmental and cultural factors as direct predictors of the distribution of languages we can more clearly understand the processes by which this distribution has emerged.

To test the relative importance of the hypotheses outlined above we used a geographical information system (GIS) to construct a database that integrates ethnographic, environmental, and language data. Data on languages, their geographical distributions, and number of speakers were derived from the fifteenth edition of the *Ethnologue* (1). (Digital language maps are produced by Global Mapping International http://www.gmi.org.) The *Ethnologue* represents the most comprehensive cataloguing of the world's 6,912 known languages, with mutual intelligibility between speakers of different varieties being the main criterion used to determine what constitutes a distinct language. As many of the native languages of the Americas and Australia show restricted ranges because of their colonial history and population replacement, languages from these

Table 1. Spearman rank correlations between language area and environmental variables

	NPP	MGS	SPDV	ALT	LAT	PC1	PC2
Area	−0.011	**−0.356**	**−0.371**	**0.237**	**0.176**	**−0.302**	**−0.106**
NPP		**0.528**	**0.155**	−0.009	**−0.539**	**0.502**	**0.426**
MGS	**0.528**		**0.638**	0.016	**−0.609**	**0.931**	**0.175**
SPDV	**0.155**	**0.638**		**0.304**	**−0.192**	**0.760**	**−0.306**
ALT	−0.009	0.016	**0.304**		**0.223**	**0.205**	**−0.631**
LAT	**−0.539**	**−0.609**	**−0.192**	**0.223**		**−0.532**	**−0.545**
PC1	**0.502**	**0.931**	**0.760**	**0.205**	**−0.532**		−0.010
PC2	**0.426**	**0.175**	**−0.306**	**−0.631**	**−0.545**	−0.010	

All correlations in boldface type significant at $P < 0.01$, otherwise not significant at $P = 0.05$. Area, log language area; NPP, net primary productivity; MGS, mean growing season; SPDV, plant species diversity; ALT, standard deviation of altitude; LAT, absolute latitude; PC1, principal component 1; PC2, principal component 2.

regions are not included in the analyses. Also languages that are only found on islands without any other languages (as is the case with many of the languages found on the Pacific islands) were excluded, leaving 4,233 languages. We used the GIS software package ArcGIS (version 9.1) to calculate the longitudinal and latitudinal location of the midpoint of each language polygon and the area covered by each polygon. Language areas are heavily positively skewed (i.e., the vast majority of languages cover quite small areas, but some have vast ranges) so they were log transformed and all of the data presented here refer to the $\log_{10}$ of language area.

The *Ethnologue* contains limited ethnographic information so where possible the language spoken by societies in the *Ethnographic Atlas* (EA) (33) was identified and the ethnographic information was appended to the language database. There are 602 societies for which their native language could be reliably matched to an entry in the *Ethnologue* (supporting information (SI) Fig. S1). Societies are coded as having 1 of 5 levels of political complexity and of adopting 1 of 5 subsistence strategies (see *Methods*).

By overlaying the language map onto maps of environmental variables (see *Methods* and Fig. S2) we calculated several summary statistics for each environmental map over the entire geographical extent of any given language. For example, it is possible to calculate such things as the mean monthly precipitation in any particular month, the range of annual mean temperatures within language areas, and the minimum value for the amount of precipitation in the wettest quarter of the year. Our database contains more than 800 potentially informative environmental variables. In this study we use mean values across the language polygons for net primary productivity (NPP) as a measure of environmental productivity, mean growing season (MGS) to assess ecological risk, plant species diversity to index biological diversity, and the standard deviation across the language polygons for altitude as a measure of topographical heterogeneity. Environmental variables exhibit a moderate degree of intercorrelation (see Table 1). This can make parameter estimates for predictors in multivariate analyses difficult to interpret although it does not affect the overall fit of the model. The effect this has on the present analysis was assessed in 3 ways: (*i*) diagnostic tests were carried out (which suggest this is not a severe problem, see *SI Methods*); (*ii*) confirmatory analyses were conducted with different combinations of predictor variables; (*iii*) these and other environmental variables were reduced using principal components analysis to produce 2 principal components of variation (see *SI Methods* and Table S1).

Results and Discussion

Table 1 shows that language area is significantly but weakly correlated with a number of environmental variables, including MGS and species diversity. However, NPP was not found to correlate significantly with language area. The standard deviation of altitude also shows a significant correlation with language area, but the direction of this relationship is in the opposite direction to that predicted by the topography hypothesis. All of the environmental variables and principal components show significant correlations with absolute latitude and so could potentially explain the latitudinal gradient in language area.

To determine which cultural and environmental factors are the best predictors of language area, we ran a linear mixed model (LMM) with language area ($\log_{10}$ km^2) as the dependent variable, MGS, NPP, and plant species diversity were entered as covariates, with degree of political complexity and subsistence strategy entered as factors. Standard deviation of altitude was not included as preliminary analyses showed that the direction of the relationship remained in the opposite direction to that predicted. Post hoc tests were performed to assess where the significant differences lay within the factors. To take into account the fact that societies in the *Ethnographic Atlas* cannot be considered independent, suffering from what is known as Galton's problem (34), language family was modeled as a random factor with languages being nested within language families (as classified by the *Ethnologue*). If the factors outlined previously are important in determining language areas then they should remain as significant predictors while controlling for the effects of the other variables.

Our LMM was able to explain 55% of the variance in the area covered by languages (Table 2). All variables included in the model explain significant amounts of variation in language area except plant species diversity, with the degree of political complexity being the largest single predictor, accounting for about a quarter of the explained variance in language area. These slopes and intercepts of these variables were not found to vary significantly across language families. In line with our predictions, languages associated with more politically complex societies cover significantly larger areas than those of less complex societies (the only exception was that societies with no levels of

Table 2. Cultural and environmental predictors of language area

	df	*F*	*p*	Effect size	β	SE
Intercept	1	2353.8	<0.001		5.183	1.845
Pol	4	40.7	<0.001	0.13	—	—
Sub	4	7.1	<0.001	0.02	—	—
SPDV	1	2.4	0.122	—	−0.102	0.067
MGS	1	40.1	<0.001	0.03	−0.128	0.021
NPP	1	27.1	<0.001	0.03	11.687	2.279

Proportion of variation explained = 0.55. Pol, political complexity; Sub, subsistence strategy; SPDV, plant species diversity. The effect size of each of the variables entered into the model is calculated as the change in proportion of variance explained by removing that variable from the model.

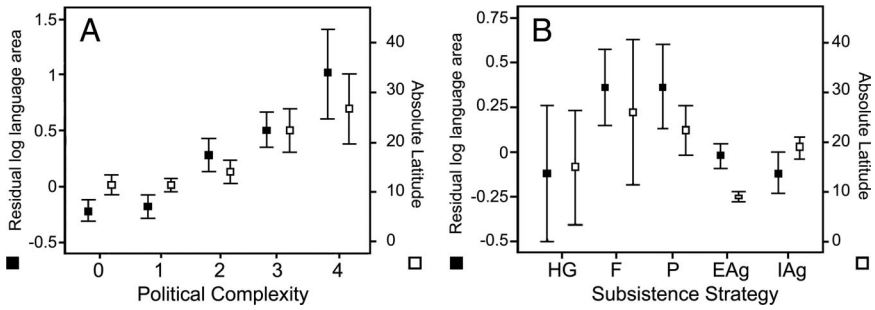


Fig. 2. Comparison between residual log language area (controlling for all other variables in Table 2) (black) and absolute latitude (white) for (*A*) levels of political complexity and (*B*) different subsistence strategies. More politically complex societies have larger language areas and are also found at higher latitudes. Subsistence strategy does not show a consistent relationship between language area and absolute latitude. HG, hunter–gatherer; F, fishing; P, pastoralism; EAg, extensive agriculture; IAg, intensive agriculture.

jurisdiction above the local level were not significantly different to those with 1 level) (Fig. 2*A* and Table S2, column *a*). This result is consistent with our hypothesis that more complex societies are better able to expand by replacing or incorporating other groups. Political complexity was also found to increase with increasing latitude (GLM: $F_4 = 15.104$, $P < 0.001$, $R^2 = 0.10$) (Fig. 2*A*), supporting the hypothesis that it is at least partly responsible for the latitudinal gradient in language diversity.

Research into the evolution of political complexity has a long and controversial history in archaeology and anthropology (35) with many contrasting explanations being proffered for the emergence of politically more complex societies. The fact that political complexity shows a latitudinal gradient provides support for hypotheses that stress environmental factors in the emergence and maintenance of such societies, in contrast to explanations that place an emphasis on nonmaterial explanations such as ideology (36, 37). There are a number of potential reasons why political complexity may be more common at higher latitudes. First, grain crops such as rice are seasonal and grow better at higher latitudes (38). Grain is more readily stored than other types of crops and therefore may have allowed the kinds of differences in wealth that underpin complex societies to develop more easily (see *SI Methods*, Fig. S3, and Tables S3 and S4). Second, good quality agricultural land may be more circumscribed at higher latitudes (39) making the costs of moving away from a ruling authority greater than at lower latitudes where suitable land is in greater abundance. Finally, the east–west axis of Eurasia, where the majority of land in the Old World is located (25), may have made it more likely that early complex societies could expand over larger areas within more northerly latitudinal bands. These hypotheses are not necessarily mutually exclusive and other explanations are also possible.

Consistent with the hypothesis that the difference in mobility associated with different subsistence strategies affects the area a language covers, pastoralists were found to have larger language areas than agriculturalists (Fig. 2*B* and Table S2, column *b*). Subsistence strategies tend to be practiced at significantly different latitudes (GLM: $F_4 = 33.140$, $P < 0.001$, $R^2 = 0.187$); however, subsistence strategy itself cannot explain the latitudinal gradient in languages as intensive agriculturalists and pastoralists are not found at significantly different latitudes (Fig. 2*B*).

To confirm that our result was not solely contingent on the particular environmental variables entered into the model, several different LMMs were run with different combinations of predictor variables. The effect sizes of variables in these models are summarized in Table 3. The variance predicted by these models does not depend to any great extent on the environmental variables included. However, not including political complexity does substantially reduce the amount of variance explained by the model, showing that political complexity is explaining variance that is not able to be explained by these environmental variables.

In this paper we have proposed that political complexity leads to languages to be spread over larger areas. It is possible, however, that the causal arrow points in the opposite direction and that the possession of a common language may facilitate the effective joining together of different groups and creation of more complex political institutions. Although this analysis is unable to establish the direction of causation statistically, other lines of evidence suggest that if large language areas have preceded the formation of more complex societies, it is unlikely

Table 3. Proportion of variance explained by parameters in confirmatory LMMs

	Pol	Sub	SPDV	MGS	NPP	Temp	Prec	PC1	PC2	Total
1	0.12	0.02	0.01	0.03	0.02	0.01	NS	—	—	0.56
2	0.13	0.02	<0.01	0.01	—	—	—	—	—	0.52
3	0.13	0.02	—	0.04	0.03	—	—	—	—	0.55
4	0.13	0.03	0.02	—	NS	—	—	—	—	0.52
5	0.13	0.02	—	0.02	—	—	—	—	—	0.52
6	0.13	0.03	0.01	—	—	—	—	—	—	0.52
7	0.14	0.02	—	—	NS	—	—	—	—	0.50
8	0.13	0.03	—	—	—	<0.01	0.01	—	—	0.52
9	0.14	0.02	—	—	—	—	—	—	—	0.50
10	0.12	0.02	—	—	—	—	—	0.02	0.01	0.53
11	—	—	NS	0.04	0.03	—	—	—	—	0.37
12	—	—	—	—	—	—	—	0.03	0.02	0.40

Pol, political complexity; Sub, subsistence strategy; Fam, language family; SPDV, plant species diversity; MGS, mean growing season; Temp, mean annual temperature; Prec, mean annual precipitation; NPP, net primary productivity; PC1, principal component 1; PC2, principal component 2; NS, nonsignificant.

to have been the dominant process. Broadly speaking, languages become widespread by 1 of 2 means, (*i*) people speaking a particular language will expand into a new area or (*ii*) a language shift will occur in which different groups of people adopt a new language. Without the kind of cohesive force that more complex political institutions provide, groups expanding into a new region will tend to fragment. Recent work has shown that languages tend to change rapidly after such splits, with people using language as badges of group identity (40) and therefore the initial situation of linguistic homogeneity is unlikely to last for long. Historical evidence also suggests that large-scale language shifts tend to occur mainly toward the lingua francas associated with politically dominant societies (e.g., Latin and Hellenistic Greek) and then mainly in regions in which there has been a significant demographic replacement (41).

The environmental variables tested here were only able to explain a small amount of the observed variance in language area in our main LMM. Mean growing season was a weak predictor of language area, which showed a consistent effect when analyzed in combination with other variables. Although the relationship between MGS and language area appears to be robust, the assertion that ecological risk has been universally the most important determinant of language diversity (9) is not supported by these results. The productivity hypothesis received only slight support from these analyses, with NPP showing a very weak or nonsignificant relationship with language area after controlling for other environmental and cultural variables. Furthermore the direction of this relationship was in the opposite direction to that predicted. Biological diversity was also found to be only a weak or nonsignificant predictor of language area in the multivariate analyses. We therefore conclude that although regions of high biological and cultural diversity do overlap to a striking degree, it is unlikely that biological diversity has any direct effect on cultural diversity on a global scale. Understanding the true nature of the relationship between these 2 aspects of diversity is important as conservation strategies based on an erroneous conception of this relationship may lead to actions that aid neither the biological species nor the people they aim to help.

While the standard deviation of altitude was significantly correlated with language area the relationship was in the opposite direction to that predicted by the topography hypothesis. It appears that the geographical separation caused by mountainous environments is not a major factor in determining the area a language covers on a global scale. Certain areas of the world such as New Guinea and Nepal do contain mountainous regions with languages covering small areas; however, cultural separation does not require geographical isolation and humans are able to maintain cultural barriers between groups despite the flow of genes and goods over these boundaries (42). A more important reason why some mountainous areas contain smaller groups may be the greater difficulty a ruling authority would have in bringing other groups under control in environments that contain mountains or other obstacles to transport (13, 43), making the evolution of more politically complex societies less likely.

As many variables correlate with latitude, teasing apart the causal relationships from the spurious correlations is not an easy task. Recent studies (14, 15) have used correlational evidence to argue that cultural diversity is a response to pathogen stress. These studies are conducted at a very course grain of analysis and do not empirically assess the alternative explanations outlined in the literature. Our study highlights the need to make concerted efforts to assemble data at an appropriate scale and distinguish between competing hypotheses that attempt to account for the distribution of human cultural groups such as languages.

It should be noted that a lot of the variation in language area is unexplained in our model. Some errors may emerge because of inconsistencies in whether spoken variants are classified as distinct languages or dialects, a well-known issue in linguistics (7, 44). We have relied on the designation of distinct languages made by the compilers of the *Ethnologue*. Boundaries between groups, and therefore spoken variants, are often in a state of flux. Expansion into a new region by speakers of a language will initially create a situation of linguistic homogeneity across that region. Over time that region might fragment into separate languages. The point at which a judgement is made on the distinctiveness of spoken varieties in that region will clearly have an impact on the number of distinct languages we describe as being spoken there. Therefore, another potentially important factor in the distribution of cultural groups could be related to the length of time that regions have been inhabited (10, 45); however, linguistic divergence can be a relatively rapid phenomenon (40, 44) and so this may be less important on the global scale we have investigated here. Despite these considerations there is no reason to suspect that any error in the measurements used in this study is of a systematic nature.

Previous studies have assumed, either implicitly or explicitly (9, 10), that cultural diversity is in equilibrium with the environment (i.e., the cultural diversity of a region will stay approximately constant over time, given a constant environment). This may have been the case before the ''Neolithic Revolution'' when the world was populated solely by hunter–gatherers with little or no differences in political complexity and military technology (44). However, if more politically complex societies do tend to replace or integrate other groups then language diversity may not be in ecological equilibrium. It is well known that the colonial expansions in modern times have caused the loss of many indigenous languages (9), and there is a growing awareness that many languages today are under threat of extinction from the effects of global socioeconomic change and the incorporation of speakers of rare languages into the economically more dominant (20). Historical evidence also suggests that the number of autonomous political units has been decreasing over the last 3,000 years (46) and the largest such units have increased in geographical extent over this time (47). Declines in language diversity, therefore, may not be just a recent phenomenon, but instead may have begun soon after the emergence of more politically complex societies.

In summary, our results show that several cultural and ecological factors are associated with the area a language covers and, by extension, the pattern of distribution of the world's languages. These results suggest that the direct effects of the cultural factors included in this analysis explain as much or more of the variation in current patterns of cultural group diversity than do the direct effects of environmental factors. In particular we have shown that the largest single factor predicting the area over which a language is spoken is the degree of political complexity exhibited by the society speaking that language. This is consistent with the hypothesis that more complex societies replace or incorporate less complex groups and thus spread their languages over larger areas. As political complexity is a property of groups, and competition often occurs between groups, rather than just between individuals, if more politically complex groups tend to replace or incorporate others, then the proportion of more politically complex societies will tend to increase over time. Such a mechanism represents a process of cultural group selection (21, 48). An interesting area for future research will be to assess the impact this process has on the biological fitness of individuals within groups (49). Increasing political complexity is almost always associated with greater degrees of social stratification, and wealth in the form of tax or tribute is often extracted by political elites from those lower down the social order (24), which could clearly have significant reproductive consequences for individuals at different levels in such societies. It will be important to assess empirically whether these costs are outweighed by benefits gained from being a member of such a group and from the advantage held in competition between groups.

Methods

Ethnographic Data. The EA variable ''levels of jurisdictional hierarchy beyond the local community'' was used to assess the degree of political complexity (the 5 categories can be thought of as 0 = no political authority beyond the local community, 1 = simple chiefdom, 2 = complex chiefdom, 3 = state, and 4 = large state). ''Subsistence economy'' was used to index the subsistence strategy used by societies. The 5 categories for subsistence strategy are: foraging (hunting and gathering, EA codes 1 and 3 combined), fishing, pastoralism, extensive agriculture (''horticulture,'' EA codes 5 and 6 combined), and intensive agriculture.

Climatic Data. High spatial resolution (30 arc secs) climatic data global raster maps have been produced by interpolating records from more than 20,000 weather stations over the period of 1950–2000 (50). These maps contain information on altitude, monthly temperature, monthly precipitation, and bioclimatic variables (e.g., mean annual temperature, precipitation in warmest quarter) (http://www.worldclim.org). Using these data we also calculated a raster map of MGS, following Nettle (9) as the number of months of the year in which the mean temperature is above 6 °C and the total precipitation in millimetres is more than twice the mean temperature in centigrade. NPP is a measure of the net amount of plant biomass converted from solar energy during photosynthesis. Data on NPP was taken from Imhoff et al. (51, 52). To reduce skewness the variables temperature seasonality, precipitation seasonality, and standard deviation of altitude were $\log_{10}$ transformed before being used in analyses. As mean annual temperature was heavily negatively skewed, it was first multiplied by -1, then had 300 added to each value to make all values positive before being $\log_{10}$ transformed and finally multiplied by -1.

Biological Diversity. The World Wildlife Fund's (WWF) global map of terrestrial ecosystems (53) provided a starting point for assessing the world's biological diversity. The digital map itself is made up of more than 14,000 distinct polygons that can be assigned to 867 terrestrial ecoregions. Following Stepp et al. (11) data on plant species diversity were taken from calculations by Kier et al. (54) on the number of vascular plants in each of the world's major ecoregions. This information was then joined to the WWF map in ArcGIS. As the terrestrial ecosystems differ in the area they cover the measure of diversity we calculated was the species density of each ecoregion (number of species in a ecoregion/area of ecoregion) and produced a raster map of global plant species density. The data were log transformed to reduce skewness.

All statistical analyses were performed using SPSS v12.0.

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